



Pro 30f/30f+ Wide-format Flatbed Printer



Maintenance Guide

Document ID: OMM-00175-H

All trademarks, registered trademarks, and product names used in this document are the property of their respective owners.

The information contained in this document is confidential and proprietary to Electronics for Imaging, Inc (EFI). This information is provided only to authorized representatives of EFI and EFI customers solely for the purpose of facilitating the use of EFI's products. No information contained herein may be disclosed to any unauthorized person for any purpose whatsoever without the prior written consent of EFI.

EFI makes no representations or warranties with respect to the contents of this document. Further, EFI reserves the right to revise or change this publication and the products it describes without notice.

English is the original language of this document.

This product may be covered by one or more of the following U.S. Patents

5,109,241, 5,150,454, 5,170,182, 5,212,546, 5,260,878, 5,276,490, 5,278,599, 5,335,040, 5,343,311, 5,398,107, 5,424,754, 5,442,429, 5,459,560, 5,467,446, 5,506,946, 5,517,334, 5,537,516, 5,543,940, 5,553,200, 5,563,689, 5,565,960, 5,583,623, 5,596,416, 5,615,314, 5,619,624, 5,625,712, 5,640,228, 5,666,436, 5,682,421, 5,729,665, 5,745,657, 5,760,913, 5,799,232, 5,818,645, 5,835,788, 5,859,711, 5,867,179, 5,937,153, 5,940,186, 5,959,867, 5,970,174, 5,982,937, 5,995,724, 6,002,795, 6,025,922, 6,035,103, 6,041,200, 6,065,041, 6,081,281, 6,112,665, 6,116,707, 6,122,407, 6,134,018, 6,141,120, 6,166,821, 6,173,286, 6,185,335, 6,201,614, 6,209,010, 6,215,562, 6,219,155, 6,219,659, 6,222,641, 6,224,048, 6,225,974, 6,226,419, 6,238,105, 6,239,895, 6,256,108, 6,269,190, 6,271,937, 6,278,901, 6,279,009, 6,289,122, 6,292,270, 6,299,063, 6,310,697, 6,321,133, 6,327,047, 6,327,050, 6,327,052, 6,330,071, 6,330,363, 6,331,899, 6,337,746, 6,340,975, 6,341,017, 6,341,018, 6,341,307, 6,347,256, 6,348,978, 6,356,359, 6,366,918, 6,369,895, 6,381,036, 6,400,443, 6,429,949, 6,449,393, 6,457,823, 6,476,927, 6,487,568, 6,490,696, 6,501,565, 6,519,053, 6,539,323, 6,543,871, 6,546,364, 6,549,294, 6,549,300, 6,550,991, 6,552,815, 6,559,958, 6,572,293, 6,590,676, 6,599,325, 6,606,165, 6,616,355, 6,618,157, 6,633,396, 6,636,326, 6,637,958, 6,643,317, 6,647,149, 6,657,741, 6,660,103, 6,662,199, 6,678,068, 6,679,640, 6,687,016, 6,707,563, 6,741,262, 6,748,471, 6,753,845, 6,757,436, 6,757,440, 6,778,700, 6,781,596, 6,786,578, 6,816,276, 6,825,943, 6,832,865, 6,836,342, 6,850,335, 6,856,428, 6,857,803, 6,859,832, 6,866,434, 6,874,860, 6,879,409, 6,885,477, 6,888,644, 6,905,189, 6,930,795, 6,950,110, 6,956,966, 6,962,449, 6,967,728, 6,974,269, 6,977,752, 6,978,299, 6,992,792, 7,002,700, 7,023,570, 7,027,187, 7,027,655, 7,031,015, 7,046,391, 7,054,015, 7,058,231, 7,064,153, 7,073,901, 7,081,969, 7,090,327, 7,093,046, 7,095,518, 7,095,528, 7,097,369, 7,099,027, 7,105,585, 7,116,444, 7,177,045, 7,177,049, 7,177,472, 7,204,484, 7,206,082, 7,212,312, 7,229,225, 7,233,397, 7,233,409, 7,239,403, 7,245,400, 7,248,752, 7,259,768, 7,259,893, 7,280,090, 7,296,157, 7,301,665, 7,301,667, 7,301,671, 7,302,095, 7,302,103, 7,304,753, 7,307,761, 7,342,686, 7,343,438, 7,349,124, 7,365,105, 7,367,060, 7,367,559, 7,389,452, 7,396,119, 7,396,864, 7,397,583, 7,397,961, 7,426,033, 7,431,436, 7,433,078, 7,453,596, 7,460,265, 7,460,721, 7,461,377, 7,463,374, 7,466,441, RE36,947, RE38,732, D341,131, D406,117, D416,550, D417,864, D419,185, D426,206, D426,206, D439,851, D444,793.

This document is published Online at <http://inkjet.support.efi.com>.

Revision History

Ver.	Date	Description	Request
A	09/09/2019	Initial release	JD email
B	09/26/2019	New style and edits	DR 5906
C	02/07/2020	Corrected filter referenced in 6.2 Replace Ink Pump Pall Filter	DR 6164
D	08/04/2021	Rotated 4-8 Carriage Height Solenoid	DR 6710
E	10/19/2022	Updated steps for 5.1 Clean and Lubricate Gantry Lead Screws	DR 7138
F	11/30/2022	Updated Required Items in 3.9 Clean Cure Lamp Lenses	DR 7200
G	11/11/2024	Updated Pro 30f+. Removed purge and sweeping with button (which does not sweep). Added 3.11 Conduct Daily White Ink Print per Oliver's 08212024 notes. Added Microlube GL 261 45086492 in place of red grease in 4.1 Clean and Lubricate Carriage Rails Changed the process for x-Rail lubrication to match the 30h	DR 7702 DR 7784 DR 7825
H	4/16/2025	Build source file in InDesign, added maintenance video links, and made updates for translation.	Glen email

Table of Contents

1.0	Introduction	2	3.8	Clean Print Heads	20
1.1	Welcome	2	3.9	Clean Cure Lamp Lenses	22
1.2	Before You Begin	2	3.10	Clean the Cyclonic Vacuum System	23
1.3	Safety	3	3.11	Conduct Daily White Ink Print	24
1.4	Customer Support Website	4	4.0	Every 40 Hours	26
1.5	Performing Maintenance	5	4.1	Clean and Lubricate Carriage Rails	26
1.6	Printer Maintenance Log	5	4.2	Clean Linear Encoder	28
1.7	Safety Labels - Printer Beam	6	4.3	Clean Carriage Height Solenoid	29
1.8	Fluid Emergency	6	4.4	Replace Cure Lamp Filters	30
1.9	First Aid - Inks and Solvents	7	4.5	Clean Anti-Static Bars	31
2.0	Scheduled Maintenance	9	4.6	Clean Table	32
2.1	Importance of Maintenance	9	5.0	Monthly	34
2.2	8 Hour Daily Production	9	5.1	Clean and Lubricate Gantry Lead Screws	34
3.0	Every 8 Hours	11	5.2	Inspect Cyclonic Vacuum System	34
3.1	Shift Change Procedures	11	6.0	Semi-Annually	38
3.2	Purge and Sweep Print Heads	12	6.1	Replace Ink Filters in Cabinet	38
3.3	Perform Nozzle Checks	13	6.2	Replace Negative Pressure System Filters	40
3.4	Check Ink Supply Levels	14	6.3	Clean E-Chains and E-Chain Trough	41
3.5	Check Ink Waste Level	15	7.0	Required Items List	43
3.6	Clean Purge Tray and Vacuum Knife	16		Pro 30f-30f+ Maintenance Log	45
3.7	Clean Print Head Plate	18			

Table of Figures

Figure	Title	Page	Figure	Title	Page
1-1	Magnetic Drive	6	3-26	Move Carriage Up	22
3-1	Purge All	12	3-27	Wipe Cure Lamp Lenses	22
3-2	Run a Nozzle Check	13	3-28	Remove Cured Ink	22
3-3	Nozzle Check Print Example	13	3-29	Liquid Separator	23
3-4	Nozzle Check Test Print (Circles indicate Dropout)	13	4-1	Move Carriage Right	26
3-5	Ink Levels	14	4-2	Carriage Rails	26
3-6	Open Left End Door	14	4-3	Wipe Excess Grease	26
3-7	Red Ink Level	14	4-4	Move Carriage Left	27
3-8	Access Waste Cabinet	15	4-5	Remove Grease from Carriage Bearings	27
3-9	Waste Bottle Indicator	15	4-6	Move Carriage Right	28
3-10	Waste Bottle Cap	15	4-7	Cleaning Linear Encoder Strip	28
3-11	Move Carriage Right	16	4-8	Move Carriage Up	29
3-12	Wipe with Lint-free Cloth	16	4-9	Solenoid Location	29
3-13	Pour Flush on Tray	17	4-10	Cleaning Solenoid	29
3-14	Re-wipe Tray	17	4-11	Cure Lamp Filter	30
3-15	Clean Vacuum Knife	17	4-12	Move Carriage Up	31
3-16	Move Carriage Up	18	4-13	Anti-static Bars	31
3-17	Rear Section of Print Head Plate	18	4-14	Cleaning Anti-Static Bars	31
3-18	Side Sections of Print Head Plate	19	4-15	Cleaning Table	32
3-19	Front Section of Print Head Plate	19	4-16	Metal Scraper	32
3-20	Move Carriage Up	20	5-1	Access Waste Cabinet	34
3-21	Flush-Saturated Lint-free Cloth	20	5-2	Vacuum Canister	34
3-22	Wipe Rear-to-Front, Single-Pass	20	5-3	Vacuum Canister Retaining Clip	35
3-23	Do Not Rub Print Head or Wipe Back-and-Forth	21	5-4	Vacuum Filter Installation	35
3-24	Do Not Wipe Multiple Heads with a Single Wipe	21	5-5	Remove Motor Filter Cover	36
3-25	Wipe Flush Residue	21	5-6	Replace Motor Filter	36

Table of Figures

Figure	Title	Page	Figure	Title	Page
6-1	Hemostat Clamping Locations	38	6-4	Filter Arrow Direction	40
6-2	Carriage Negative Pressure Filters	39	6-5	E-Chain Cleaning	41
6-3	Carriage Secondary Ink Filters	40			

Maintenance Guide

1.0 Introduction

1.0 Introduction

The guide provides instruction for certified operators to perform the required routine maintenance tasks. Download the latest version of this document at <https://inkjet.support.efi.com>.

1.1 Welcome

Welcome to the EFI Printer Maintenance Guide. Please take the time to get well acquainted with your printer by reading this document and the corresponding Operations Guide. The more you know and understand about your printer, the greater safety and production value you will derive from it.

Additional information is available in separate documents, which can be found here: <https://inkjet.support.efi.com>.

For more information on EFI and its products, visit: www.efi.com.

1.2 Before You Begin

Before performing any scheduled maintenance procedure, operators are required to complete Operator Certification Training. For more information, visit: <https://inkjet.support.efi.com/training/>

Ensure all required items are on-hand before beginning any maintenance procedure. Required items can be found in the [Required Items List](#) section of this document.

Note: You must read and understand the following documents prior to operating or servicing this printer.

- <http://inkjet.support.efi.com/doc.php?doc=683> - Inkjet Solutions Printer Safety Guide
- <http://inkjet.support.efi.com/doc.php?doc=4901> - Pro 30f+ Operations Guide
- <http://inkjet.support.efi.com/doc.php?doc=4893> - Pro 30f-30f+ Maintenance Guide (this document)

These documents are available at <http://inkjet.support.efi.com>.

1.3 Safety

Use caution when performing maintenance tasks on your printer. Adhere to all safety precautions described in the [Inkjet Printer Safety Guide](#). Make sure nothing gets caught in moving parts. Do not perform maintenance procedures while the printer is in production. Keep all open flames and other burning materials (such as cigarettes) away from the printer. Safety comments and instructions should be read and observed to avoid personal injury and/or printer damage. These comments are highlighted as follows:

WARNING:

- Comments in a red highlighted box with triangle symbol
- Used when failure to follow will result in operator injury or printer damage

Example:



WARNING: Contact between the moving carriage and a foreign object can cause injury and damage.

CAUTION:

- Comments in a yellow highlighted box
- Used when failure to follow could result in operator injury or printer damage

Example:



CAUTION: Do not apply excessive pressure. Do not try to insert the corners of the wipes into the print head nozzles.

Note:

- Comments in italics
- Used when failure to follow could result in printer errors
- Used when describing helpful tips

Example:

Note: The second wiping must be with a dry wipe to prevent print head cleaning fluid from potentially entering the print head.

1.4 Customer Support Website

	North & South Americas		Europe, Middle East, Africa	Asian Pacific (APAC)
	Customer Care			
	U.S.	855-EFI-4HLP (855-334-4457)	+31 20 658 8070	+1 650 357 4790
		inkjet.support@efi.com	EuroInk@efi.com	
	Canada Mexico South America	+1 650 357 4790		
	Technical Support - https://inkjet.support.efi.com			
	U.S. Canada	855-EFI-4HLP (855-334-4457)	+32 2 749 94 50	+65 6221 2765
		inkjet.support@efi.com		
	Mexico South America	+1 412 690 4321	DE +49 2102 745 4500 NL +31 20 658 8080/8069 UK +44 12462 98085	
	Order Parts			
	U.S.	Inkjet.Orders@efi.com	EuroParts@efi.com	InternationalOrders@efi.com
	Canada Mexico South America	InternationalOrders@efi.com		

1.5 Performing Maintenance

Most efficient maintenance routine:

- Read each maintenance task fully before starting.
- Verify that you have all the materials required to complete each maintenance procedure.
- Perform all maintenance tasks as scheduled in the maintenance log.
- Perform maintenance activities either before or after printing production shifts.

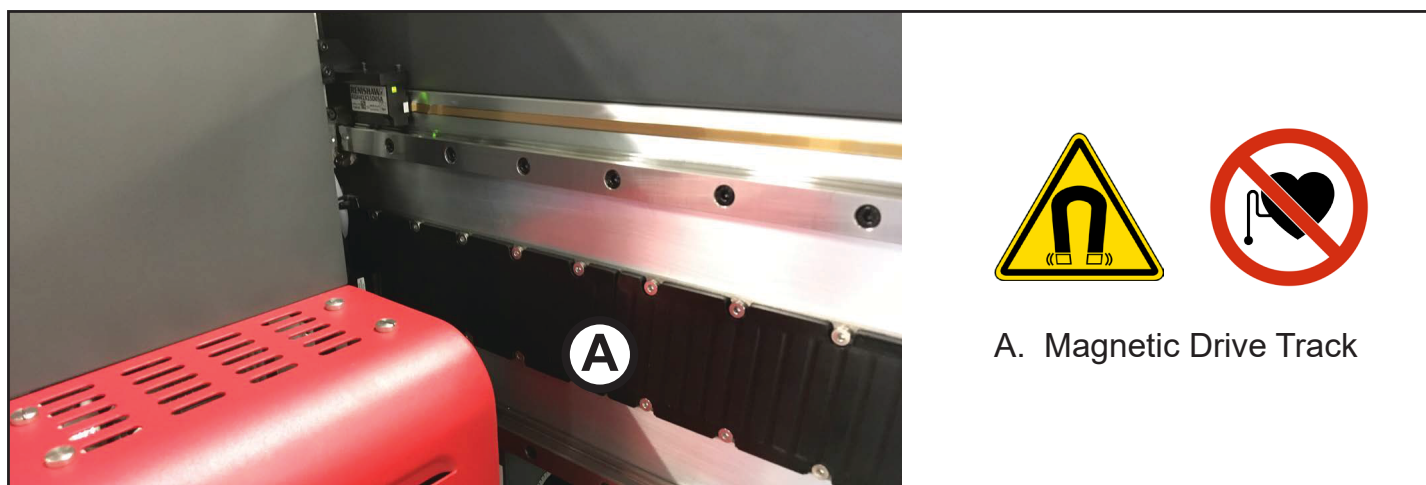
1.6 Printer Maintenance Log

The [Pro 30f-30f+ Maintenance Log](#) is at the end of this document.

The Printer Maintenance Log lists the maintenance tasks, as well as the means for tracking the completion of each item. Print the first side once per week for daily and weekly tasks, and print the second page once per year for less frequent maintenance. When filling out the Maintenance Log, record the current date and time (if applicable), and write your initials in the space provided. Please retain copies of the Maintenance Log. Field Engineers will often ask to review your maintenance records during service visits.

1.7 Safety Labels - Printer Beam

The Printer Beam contains powerful magnets that assist the carriage movement. Persons with implanted medical devices must take extreme caution to avoid being within or contacting these magnets. Operators and Technicians must take extreme caution when using tools or other metal objects within 24" (61 cm) of the Printer Beam Magnets.



1-1 Magnetic Drive

1.8 Fluid Emergency

If an emergency situation involving inks or maintenance fluids arises, contact local emergency services and download any SDS (Safety Data Sheets) and make available to emergency service personnel.

SDS are available on the Inkjet Support Site, <https://inkjet.support.efi.com/main.php> under Documents > SDS (Safety Data Sheets).

1.9 First Aid - Inks and Solvents

Follow first aid directions based on these situations.

Eye contact

- Immediately flush with cool water (≥ 15 minutes)
- Seek medical attention

Ingestion

- Do not induce vomiting; drink two glasses of water
- Seek medical attention

Skin contact

- Remove contaminated clothing
- Flush with plenty of cool water and non-abrasive soap
- Seek medical attention if irritation develops

Inhalation

- Move person to fresh air
- Seek medical attention if symptoms develop

Maintenance Guide

2.0 Scheduled Maintenance

2.0 Scheduled Maintenance

Scheduled maintenance is essential to promoting greater printer reliability, increased print quality, and greater production results. Use only recommended fluids, lubricants and parts, as described in EFI documents.

2.1 Importance of Maintenance

Routine maintenance can help to:

- Maintain printer up time
- Maximize productivity
- Maintain peak image quality

To help you service your printer, we provide a Maintenance Log which makes tracking routine maintenance easy. The Maintenance Log provides you with a list of preventative maintenance tasks, as well as the means for tracking the completion of each task. Please keep your maintenance log up to date. Your EFI Service Engineer will review your maintenance log during scheduled visits.

2.2 8 Hour Daily Production

The maintenance guidelines are based on an eight hour print production schedule, five days per week. Adjust your maintenance intervals if your operating shifts are longer than eight hours, five days per week.

Maintenance Guide

3.0 Every 8 Hours

3.0 Every 8 Hours

The following maintenance tasks must be performed every 8 hours of production.

3.1 Shift Change Procedures

Certain tasks must be performed at printer start up, at the start of each new shift, and/or at the end of each shift (or after 8 hours of printer operation).

The procedures for Start of Shift include:

- Conduct a Purge and Sweep
- Run a Nozzle Check. If your nozzle check indicates that a nozzle that isn't flowing properly, conduct a print head cleaning. If the nozzle check looks good, print heads should not be cleaned at the beginning of a shift because they will have been cleaned at the end of the previous shift.
- Check Ink Supply Levels
- Empty the Ink Waste Container
- Clean the Purge Tray and Vacuum Knife
- Clean Carriage Print Plate
- Emptying the Cyclonic Vacuum System

Procedures for the End of each Shift (or after each 8 hours of printer operation) include:

- Conduct a Purge and Sweep
- Clean the Print Plate
- Clean the Print Heads
- Clean Lamp Lenses and Fan Shrouds

Each of these procedures is described in the sections below.

3.2 Purge and Sweep Print Heads

Periodically a purge of the print heads may be required. At a minimum, the print heads should be purged and a head sweep should be performed at start up and every 8 hours of operation.

PPE Items:

NITRILE GLOVES DISPOSABLE
SAFETY GLASSES

How-to-Video:



[CLICK TO PLAY](#)

Directions:

- From the printer software's toolbar, choose the Purge All icon.
(The gear symbol offers options to purge white, color, or all heads.)
The system automatically follows this up with a head sweep.

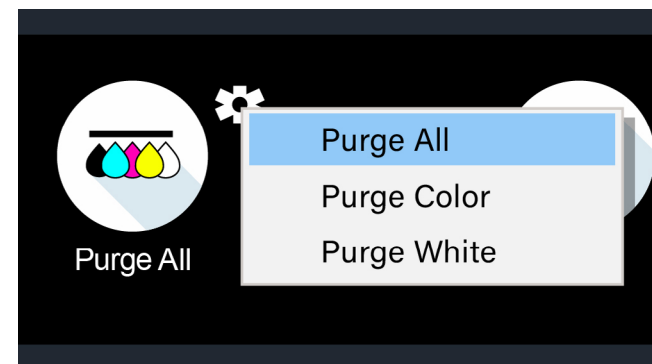


Figure 3-1 Purge All

3.3 Perform Nozzle Checks

The Nozzle Check test provides a test print that demonstrates whether all print heads are firing properly. Perform a Nozzle Check prior to beginning normal printer operations, in order to ensure that all print heads are firing properly.

PPE Items:

NITRILE GLOVES DISPOSABLE, SAFETY GLASSES

Directions:

1. Load media.
2. Click the Gear icon to open the Media Thickness pop-up; verify X and Y Position.
3. On the main GUI window, click the Measure Thickness button ([Figure 3-2, A](#)).
4. Click the Run Test Gear and verify the X and Y position.
Then click the Measure Media Button. ([Figure 3-2, B](#))
5. Examine the nozzle check printout.
6. If any nozzles are missing, perform the purge-and-wipe process.
7. Print another nozzle check.

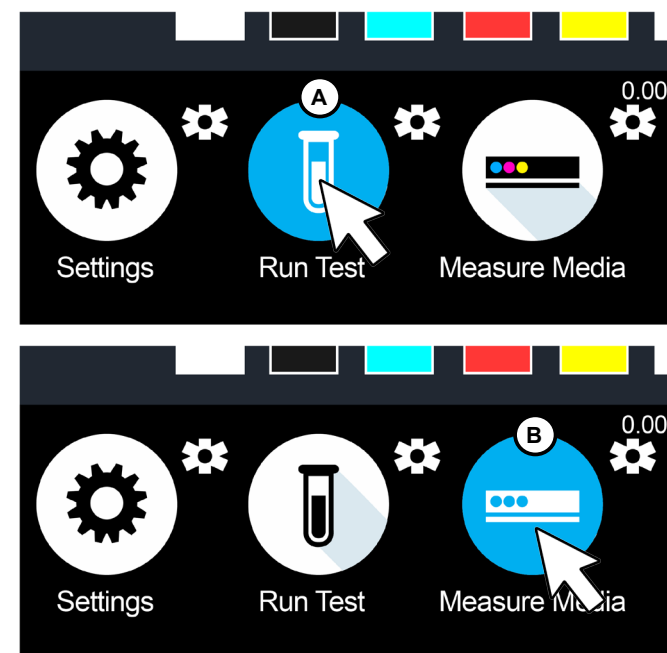


Figure 3-2 Run a Nozzle Check



Figure 3-3 Nozzle Check Print Example

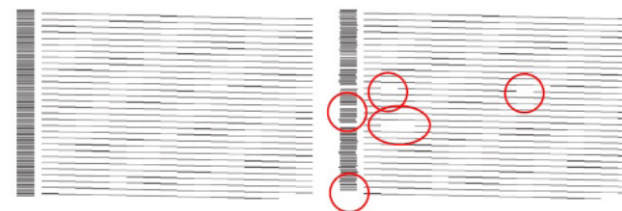


Figure 3-4 Nozzle Check Test Print (Circles indicate Dropout)

3.4 Check Ink Supply Levels

PPE Items:

NITRILE GLOVES DISPOSABLE

SAFETY GLASSES

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. Check the ink supply levels on the main screen of the printer software.



CAUTION: Always wear PPG (personal protective gear) when working with inks or fluids.

2. In the rear of the printer, open the Ink Compartment.
3. Verify the ink levels in the tanks.
4. Refer to Operations Manual for Ink Loading procedure.

Note: Remaining ink may be added to new ink if the expiration dates of two ink bottles are identical.

Note: The printer beeps and the software displays a warning (red ink level) if ink level is 10% or lower. (Figure 3-7, A)

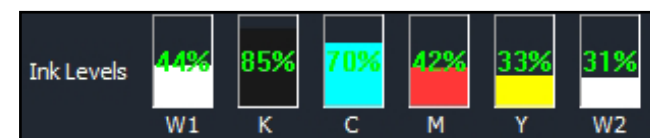


Figure 3-5 Ink Levels



Figure 3-6 Open Left End Door

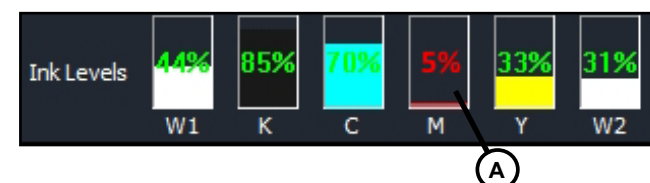


Figure 3-7 Red Ink Level

3.5 Check Ink Waste Level

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE

SAFETY GLASSES

Directions:

1. Open the door on rear of printer to access the waste cabinet ([Figure 3-8](#), A).
2. Check the waste bottle ink level. The printer beeps, and the software Waste Bottle indicator ([Figure 3-9](#), B) displays in gray if the waste ink level is above half.

If the waste bottle is more than half full:

3. Unscrew waste ink tube cap ([Figure 3-10](#), C).
4. Remove the waste ink container.
5. Dispose of waste ink within limits of local regulations.
6. Re-insert the waste ink container.
7. Screw the waste bottle tube cap back on.

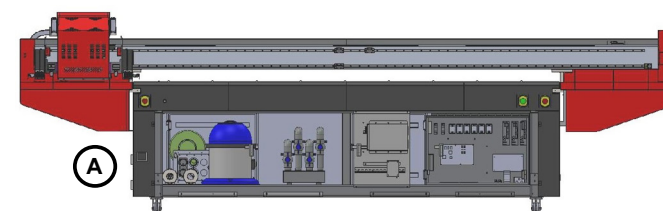


Figure 3-8 Access Waste Cabinet

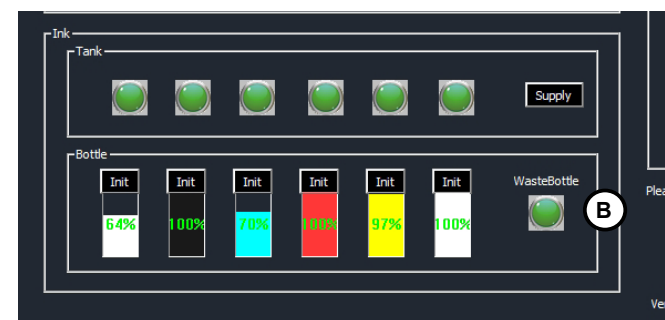


Figure 3-9 Waste Bottle Indicator



Figure 3-10 Waste Bottle Cap

3.6 Clean Purge Tray and Vacuum Knife

The purge tray and vacuum knife must be kept clean of ink waste. Light from the UV lamps cures ink in the tray and creates drainage problems if not kept clean of ink on a daily basis.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM, INK EFI WIDE FORMAT SOLVENT FLUSH

How-to-Video:

[CLICK TO PLAY](#)

Directions:

1. From the software main screen, click the carriage right arrow twice ([Figure 3-11](#)).
2. Wipe tray and trough with a clean, lint-free cloth to remove most ink.

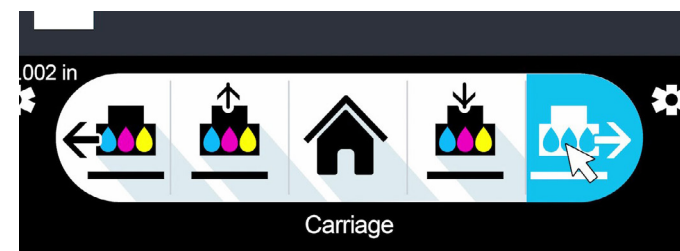


Figure 3-11 Move Carriage Right



Figure 3-12 Wipe with Lint-free Cloth

3. Pour 1-2 ounces of flush into purge tray and trough.
4. Use a clean, lint-free cloth to wipe flush and remaining ink toward drain hole
5. Use a clean, lint-free cloth, saturated with flush, to clean vacuum knife.
6. Use clean, lint-free cloth to clean remaining ink from purge tray, trough and vacuum knife to remove any remaining ink and flush residue..

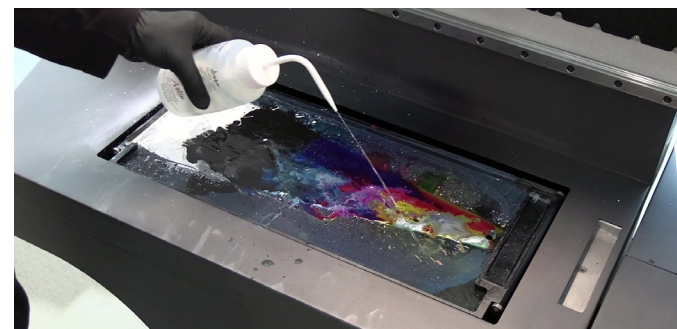


Figure 3-13 Pour Flush on Tray



Figure 3-14 Re-wipe Tray

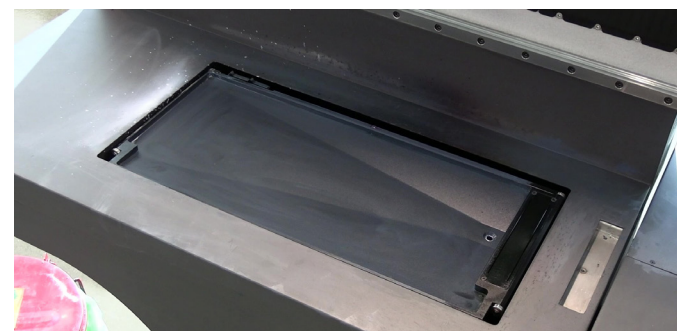


Figure 3-15 Clean Vacuum Knife

3.7 Clean Print Head Plate

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE
SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM
INK EFI WIDE FORMAT SOLVENT FLUSH

How-to-Video:

[CLICK TO PLAY](#)

Directions:

1. From the printer software, click the carriage up arrow to raise the print carriage and allow access to the print head plate ([Figure 3-16](#)).
2. Using a clean, lint-free cloth, saturated with flush, wipe across the rear section of the print head plate.

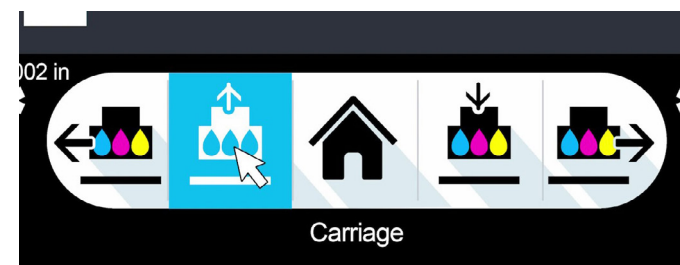


Figure 3-16 Move Carriage Up

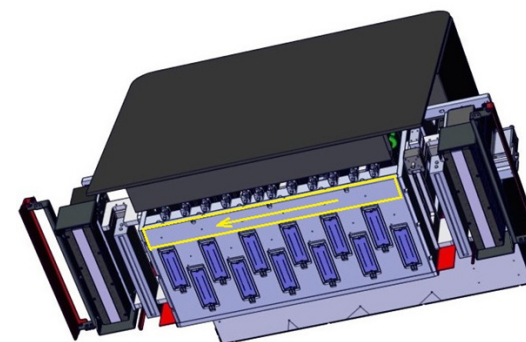


Figure 3-17 Rear Section of Print Head Plate

3. Then, wipe each side section from rear to front.
4. Finally, wipe across the front section.



CAUTION: Always wipe from rear to front. Do not wipe print heads and print head plate together with one motion.

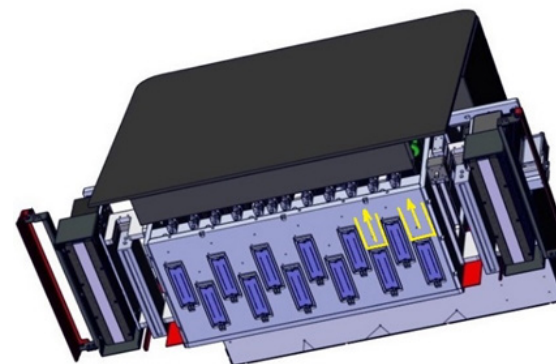


Figure 3-18 Side Sections of Print Head Plate

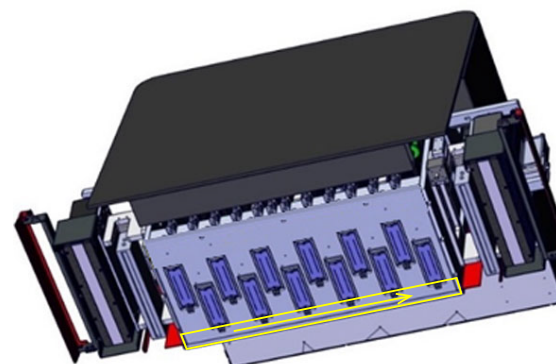


Figure 3-19 Front Section of Print Head Plate

3.8 Clean Print Heads

Note: Must be completed every four (4) hours on white ink print heads, if printing white ink.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE
SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM
INK EFI WIDE FORMAT SOLVENT FLUSH

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. From the printer software main window, click the carriage up arrow ([Figure 3-20](#)) to raise the print carriage and allow access to print heads.
2. Using a clean, lint-free cloth saturated with flush, wipe one print head at a time, from rear to front, in a single pass.

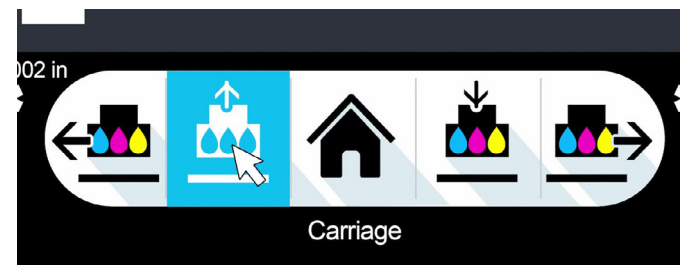


Figure 3-20 Move Carriage Up



Figure 3-21 Flush-Saturated Lint-free Cloth

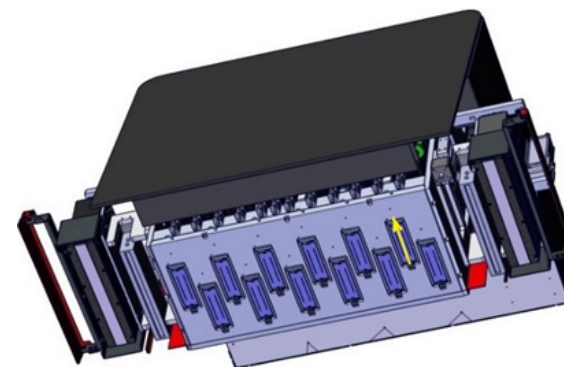


Figure 3-22 Wipe Rear-to-Front, Single-Pass



CAUTION: Do NOT RUB the print heads. If print head requires more than one wipe, use a new clean, lint-free wipe and repeat the single-wipe process. Rubbing damages the heads.

3. Repeat process on each print head.
4. Inspect print heads and print head plate to ensure all ink is removed.
5. If necessary, repeat process.

Note: Always perform a purge and sweep after wiping the print heads. (See [Section 3.2](#))

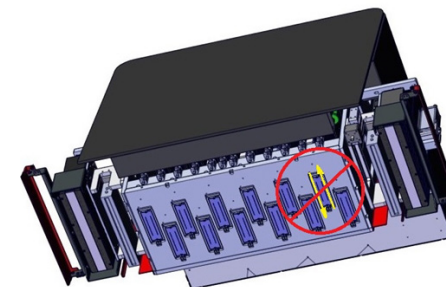


Figure 3-23 Do Not Rub Print Head or Wipe Back-and-Forth

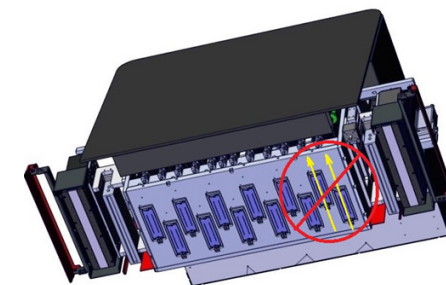


Figure 3-24 Do Not Wipe Multiple Heads with a Single Wipe



Figure 3-25 Wipe Flush Residue

3.9 Clean Cure Lamp Lenses

The printers can generate over-spray and ink mist when printing. It is essential that the lamps be cared for as described. Check and clean the lamp lenses and fan shrouds.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM, ISOPROPYL ALCOHOL (IPA)

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. From the main window, click the carriage up arrow ([Figure 3-26](#)) to raise print carriage and allow access to cure lamp lenses.
2. Using a clean, lint-free cloth, saturated with IPA (alcohol), wipe the cure lamp lenses.
3. Inspect cure lamp lenses to ensure all dust, debris, and ink have been removed.
4. If necessary, a straight edge razor may be used to remove cured ink.



WARNING: Do not use scrub pads on lamp lenses! This can result in hazing of the lenses, reducing the lamp intensity.

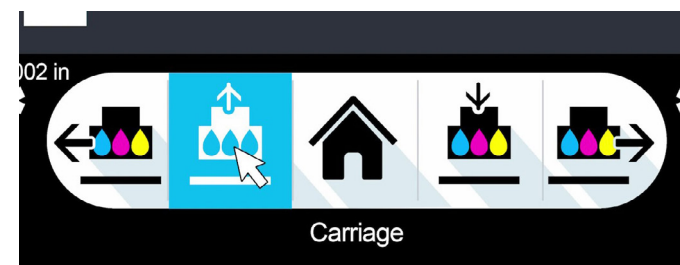


Figure 3-26 Move Carriage Up



Figure 3-27 Wipe Cure Lamp Lenses



Figure 3-28 Remove Cured Ink

3.10 Clean the Cyclonic Vacuum System

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE

SAFETY GLASSES

How-to-Video:

[CLICK TO PLAY](#)

Directions:

1. Press and hold the Liquid Separator buttons for ten (10) seconds.

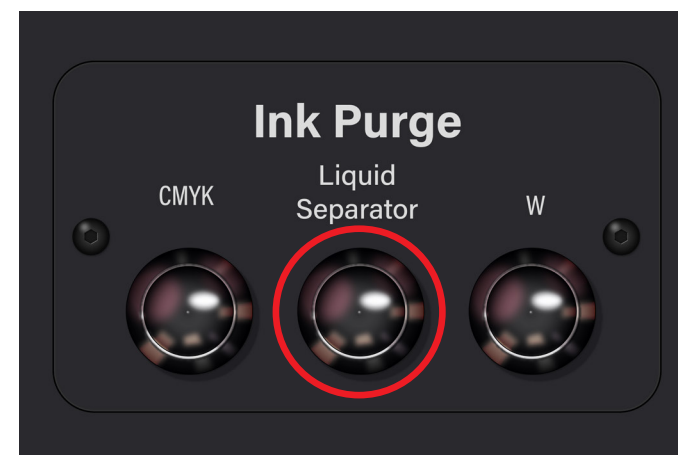


Figure 3-29 Liquid Separator

3.11 Conduct Daily White Ink Print

Printers with white ink should print white at least once each day. If your printer does not require the use of white ink during your shift, you should print a white flood of at least 1-2sqm as part of your daily maintenance.

1. Run a daily white print to keep white ink moving through the heads, prevent clogged nozzles, and increase the life of the heads.

Note: white print file can be located here:

<https://inkjet.support.efi.com/doc.php?doc=5700&id=>

Maintenance Guide

4.0 Every 40 Hours

4.0 Every 40 Hours

The following maintenance tasks must be performed every 40 hours of production.

4.1 Clean and Lubricate Carriage Rails

Weekly, clean the rails and lubricate them with Microlube. The carriage bearings pick up the grease as the carriage moves along the rails and encounters the lubrication.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM, MICROLUBE GL 261, DISPENSER, GREASE, WITH 12" FLEX HOSE, NOZZLE EXTENSION, NEEDLE, GREASE NOZZLE, 1/8 FNPT

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. From the main software screen, click the carriage right arrow ([Figure 4-1](#)) four (4) times to move the carriage and evenly spread Microlube along the carriage rails.
2. Using a clean, lint-free cloth, wipe the top and bottom of each carriage rail, removing all grease, dust and debris.
3. Lubricate rails with Microlube. Note, The grease gun and needle provided may be used alternatively to grease the interior of the bearings.
4. Using a clean, lint-free cloth, wipe any excess grease from carriage rail end.
([Figure 4-3](#))

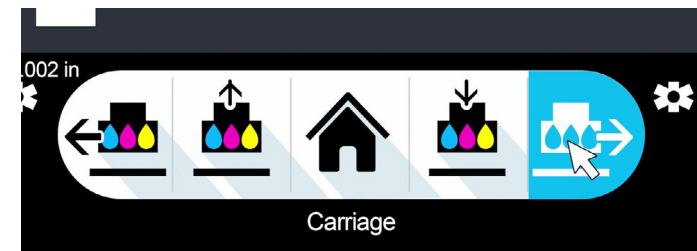


Figure 4-1 Move Carriage Right



Figure 4-2 Carriage Rails



Figure 4-3 Wipe Excess Grease

5. Click the Move Carriage Left button once to return the carriage to its home position. ([Figure 4-4](#))
6. Using a clean, lint-free cloth, wipe carriage bearings, removing excess grease. ([Figure 4-5](#))

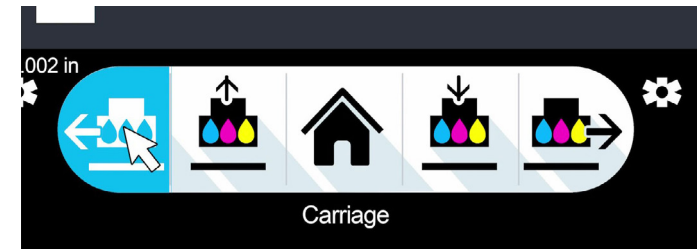


Figure 4-4 Move Carriage Left

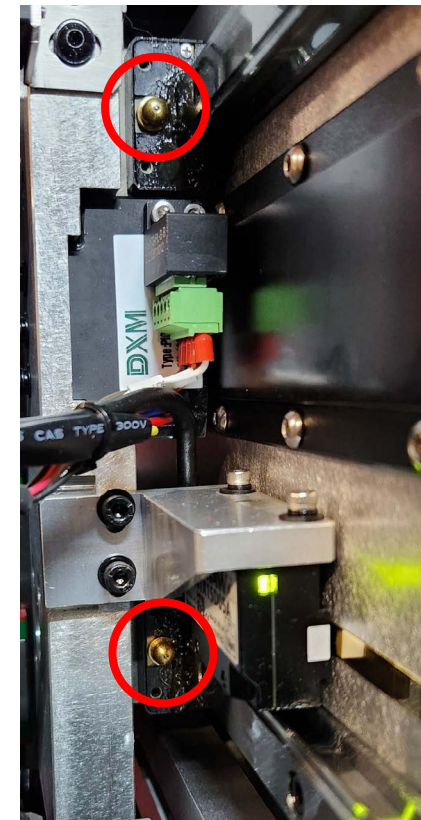


Figure 4-5 Remove Grease from Carriage Bearings

4.2 Clean Linear Encoder

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE
SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM
ISOPROPYL ALCOHOL (IPA)

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. Using a clean, lint-free cloth saturated with IPA, wipe the linear encoder strip.
2. To clean the linear encoder strip behind the print carriage, click the carriage right arrow ([Figure 4-6](#)) on the main screen. Click it a couple of times to move the carriage until it exposes the remainder of the linear encoder strip.
3. Locate the linear encoder reader.
4. Using a clean, lint-free cloth saturated with IPA (alcohol), wipe the inside surface of the linear encoder reader.

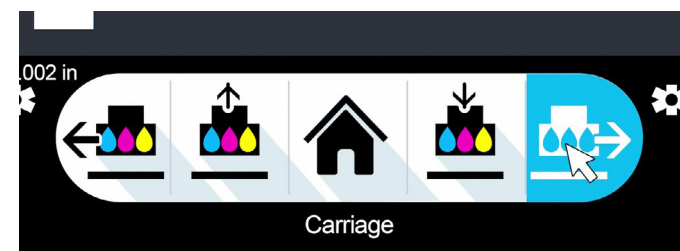


Figure 4-6 Move Carriage Right



Figure 4-7 Cleaning Linear Encoder Strip

4.3 Clean Carriage Height Solenoid

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

WIPES 9X9 CLEAN ROOM, ISOPROPYL ALCOHOL (IPA)

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. From the printer software main window, click the carriage up arrow ([Figure 4-8](#)) to raise the print carriage and allow access to the print head plate.
2. Open front carriage cover.
3. Locate the carriage height solenoid within the print carriage ([Figure 4-9](#)).
4. Push down the top of the carriage height solenoid to expose the bottom of the solenoid on the print head plate.
5. Using a clean, lint-free cloth saturated with IPA (alcohol), wipe the end of the solenoid until clean.
6. Close front carriage cover.

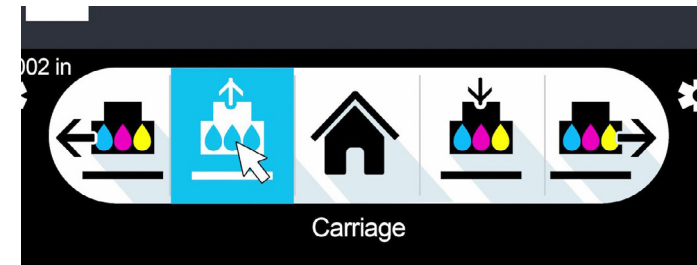


Figure 4-8 Move Carriage Up



Figure 4-9 Solenoid Location



Figure 4-10 Cleaning Solenoid

4.4 Replace Cure Lamp Filters

The printer generates over-spray and ink mist when printing. It is essential that the lamps be cared for as described. Check the filters weekly and replace them if saturated with ink or debris.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

45250905 FILTER, LAMP, FJ100 GEN 2, 225X20, 6 PAC

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. Carriage cover may be removed for easier access.
2. Remove the filter and discard it.
3. Install a new filter, ensuring that it fits properly.
4. Repeat this procedure for each lamp.



Figure 4-11 Cure Lamp Filter

4.5 Clean Anti-Static Bars

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

ISOPROPYL ALCOHOL (IPA), NYLON BRUSH

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. From the printer software main window, click the carriage up arrow ([Figure 4-12](#)) to raise the print carriage and allow access to the print head plate.
2. Power the printer OFF.
3. Using a nylon brush and IPA (alcohol), clean the anti-static bars.

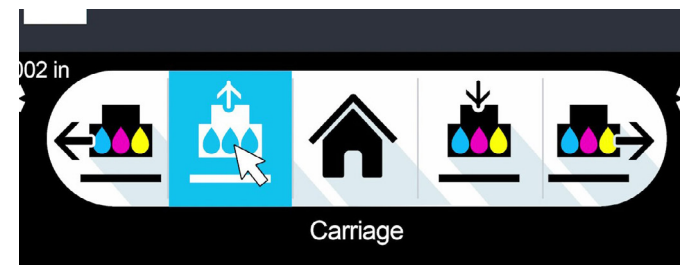


Figure 4-12 Move Carriage Up

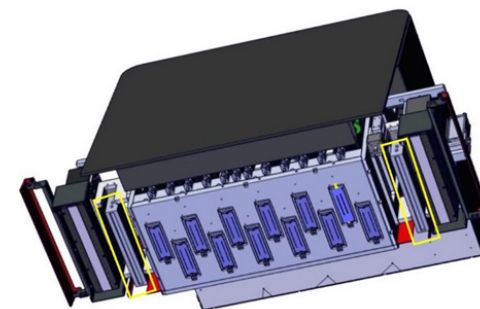


Figure 4-13 Anti-static Bars



Figure 4-14 Cleaning Anti-Static Bars

4.6 Clean Table

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

METAL SCRAPER, ISOPROPYL ALCOHOL (IPA), WIPES 9X9 CLEAN ROOM, TIP CLEANER TOOL

How-to-Video:



[CLICK TO PLAY](#)

Directions:

Caution: Always wear PPG (personal protective gear) when working with inks or fluids.

1. Move the gantry to the rear of the printer to gain access to the entire table.
2. Turn off the table vacuum.
3. Dampen a lint-free wipe with IPA and thoroughly wipe down the print table surface.
4. Using the scraper, carefully remove all ink deposits from the print table surface.
5. Using a lint-free wipe and IPA, thoroughly wipe down the print table surface again.
6. Using the tip cleaner tool or needle, make sure that each print table hole is perfectly clean and free of ink.
7. Dampen a lint-free wipe with IPA and thoroughly clean the surface of the print table again.
8. Ensure all the IPA has dried/evaporated before turning the vacuum back on.



Figure 4-15 Cleaning Table

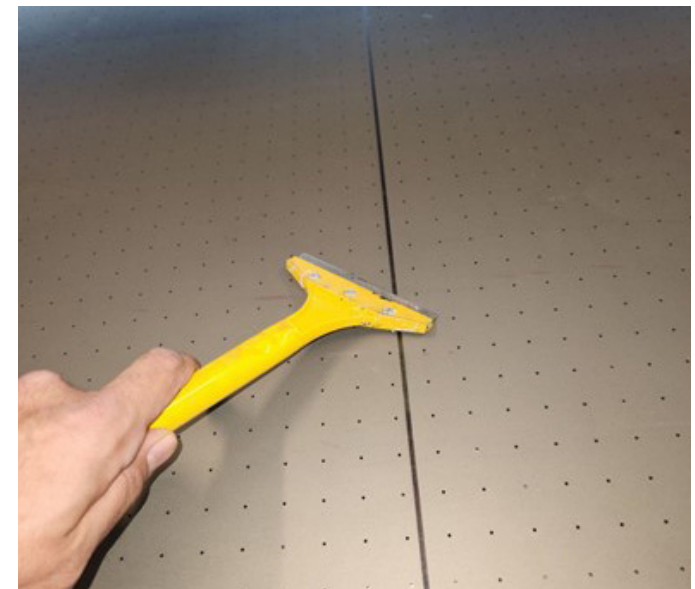


Figure 4-16 Metal Scraper

Maintenance Guide

5.0 Monthly

5.0 Monthly

The following maintenance tasks must be performed monthly.

5.1 Clean and Lubricate Gantry Lead Screws

Directions:

Contact an Field Service Engineer for Service.

5.2 Inspect Cyclonic Vacuum System

Inspect the Cyclonic Vacuum System. When white filter seal inside of vacuum canister appears discolored, replace Vacuum Canister Filters.

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. Open the left access door located in the front of the printer ([Figure 5-2](#), A).
2. Locate vacuum canister. ([Figure 5-3](#)).



Figure 5-1 Access Waste Cabinet



Figure 5-2 Vacuum Canister

3. Loosen two (2) retaining clips on vacuum canister.
4. Remove canister top.
5. Inspect white canister filter. If discolored or soiled, replace filter and clean interior of canister.



Figure 5-3 Vacuum Canister Retaining Clip



Figure 5-4 Vacuum Filter Installation

6. Remove motor filter cover and replace motor filter.
7. Replace motor filter cover and canister top, remembering to tighten retaining clips.

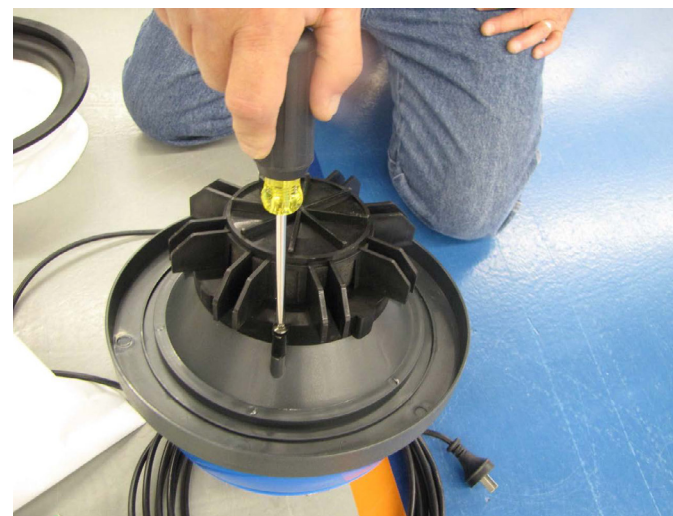


Figure 5-5 Remove Motor Filter Cover



Figure 5-6 Replace Motor Filter

Maintenance Guide

6.0 Semi-Annually

6.0 Semi-Annually

The following maintenance tasks must be performed every 6 months.

6.1 Replace Ink Filters in Cabinet

PPE Items:

GLOVES NITRILE DISPOSABLE, SAFETY GLASSES

Required Items:

HEMOSTAT CLAMPS, FILTERS, BULK INK, CAPSULE, 6 MICRON, LUER, WIPES 9X9 CLEAN ROOM

How-to-Video:



[CLICK TO PLAY](#)

Directions:



WARNING: Be sure printer is off before continuing.

1. Power the printer OFF.
2. Open the ink compartment door.



CAUTION: Be sure to turn the printer OFF for the remainder of this task.

3. Place lint-free cloths to catch any ink spillage.
4. Using EFI-provided hemostats, clamp the ink tubes ([Figure 6-1](#), A and B) above and below the ink filter.
5. Unscrew the used filter from the top, then from the bottom, and discard used filter.

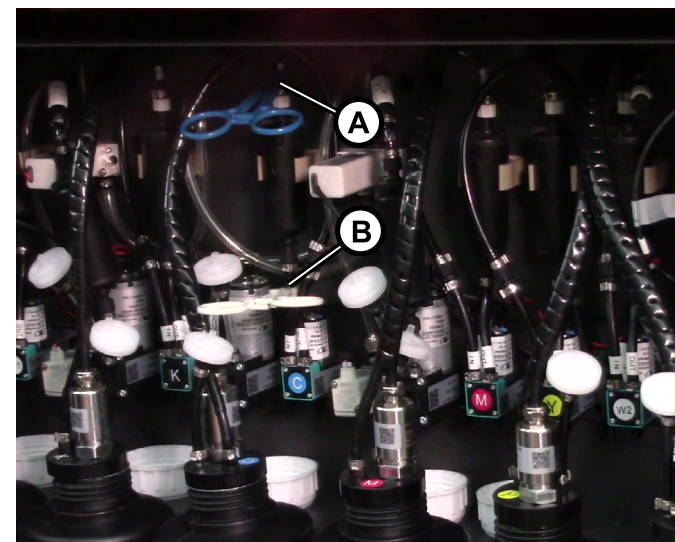


Figure 6-1 Hemostat Clamping Locations

6. Preload connector tube by gently twisting it a couple of turns prior to inserting the new filter. Release twisted connector and it automatically initiates the connection as it untwists.
7. Gently twist connector to create a firm but not over-tight connection.
8. Release clamps above and below ink filter.
9. Using a lint free cloth, wipe the ink filter and ink tubes to remove any spillage.
10. When finished, close the ink compartment door.



Figure 6-2 Carriage Negative Pressure Filters

6.2 Replace Negative Pressure System Filters

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

FILTER, BULK INK, CAPSULE, 6 MICRON, LUER, HEMOSTAT CLAMP

How-to-Video:



[CLICK TO PLAY](#)

Directions:

1. Power the printer OFF.



WARNING: Be sure printer is off before continuing.

2. Clamp the lines above the negative pressure filter.
3. Disconnect used filter and discard.
4. Ensure that arrow on new filter points down, to orient the filter in the correct direction.
5. Preload connector tube by gently twisting it a couple of times prior to inserting new filter. Release twisted connector and it automatically initiates the connection as it untwists.
6. Gently twist connector to create a firm but not over-tight connection.
7. Release clamp above filter.

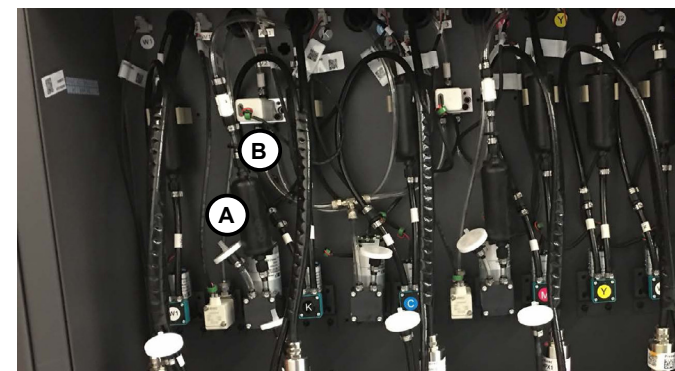


Figure 6-3 Carriage Secondary Ink Filters

A	Negative Pressure Filter
B	Clamp Ink Line Here



Filter arrow points down

Figure 6-4 Filter Arrow Direction

6.3 Clean E-Chains and E-Chain Trough

PPE Items:

GLOVES NITRILE DISPOSABLE BOX LARGE, SAFETY GLASSES

Required Items:

Shop Vacuum with Brush attachment

How-to-Video:

[CLICK TO PLAY](#)

Directions:

1. Move the carriage to the far right of the printer by clicking the **Move Carriage Right** button on the main GUI screen.
2. Use a vacuum to clean the E-Chain and E-Chain Trough.
3. Inspect the E-Chain and E-Chain Trough for Damage.
4. When completed, return the carriage to its original position by selecting the Home button on the GUI.

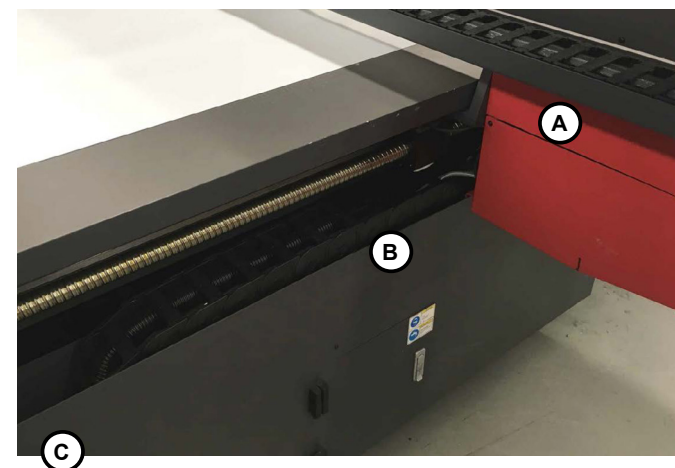


Figure 6-5 E-Chain Cleaning

A	Carriage E-Chain
B	Gantry E-Chain
C	E-Chain Trough

Maintenance Guide

7.0 Required Items List

7.0 Required Items List

This section lists all items required to perform scheduled maintenance.

DESCRIPTION	PART #	DESCRIPTION	PART #
ABRASIVE SHEET ROLL	45120360	ADDITIONAL ITEMS	NO PART #
MICROLUBE GL 261	45086492	Hemostat clamps	
FILTER, BULK INK, CAPSULE, 6 MICRON, LUER	45087559	Isopropyl alcohol (IPA)	
FILTER, LCF, HDC II 6UM, FEMALE LUER, PP WHITE INK	45080375	Razor Blade Handle	
WIPES 9X9 CLEAN ROOM	45077321	Safety Glasses	
DISPENSER, GREASE, WITH 12" FLEX HOSE, NOZZLE EXTENSION	45086491	Shop Vacuum with Brush attachment	
NEEDLE, GREASE NOZZLE, 1/8 FNPT	45087932	CLEANING AND MAINTENANCE FLUIDS	
FILTER, LAMP, FJ100 GEN 2, 225X20, 6 PAC	45250905	Ink EFI Wide Format Solvent Flush	45080261

Maintenance Guide

Pro 30f-30f+ Maintenance Log

Pro 30f-30f+ Maintenance Log

The maintenance guidelines are based on a production schedule of eight hours of printing, five days/week.
Adjust your maintenance intervals if your operating shifts are longer than eight hours of printing.

Week: ____/____/____

Serial #: _____

At Start Up and Every 8 Hours	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
1. Purge and Sweep Print Heads																					
2. Perform Nozzle Checks																					
3. Check Ink Supply Levels																					
4. Check Ink Waste Level																					
5. Clean Purge Tray and Vacuum Knife																					
6. Clean Print Head Plate																					
7. Clean Print Heads																					
8. Clean Cure Lamp Lenses																					
9. Clean the Cyclonic Vacuum System																					
10. Conduct Daily White Ink Print																					

Every 40 Hours	1	2	3
1. Clean and Lubricate Carriage Rails			
2. Clean Linear Encoder			
3. Clean Carriage Height Solenoid			
4. Replace Cure Lamp Filters			
5. Clean Anti-static Bars			
6. Clean Table			

Pro 30f-30f+ Maintenance Log

The maintenance guidelines are based on a production schedule of eight hours of printing, five days/week.
Adjust your maintenance intervals if your operating shifts are longer than eight hours of printing.

Year: _____

Serial #: _____

Monthly	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
1. Clean and Lubricate Gantry Lead Screws												
2. Inspect Cyclonic Vacuum System												

Semi-Annually	6	12
1. Replace Ink Filters in Cabinet		
2. Replace Negative Pressure System Filters		
3. Clean E-Chains and E-Chain Trough		